ESTIMATION OF BLOOD GLUCOSE BY GLUCOMETER

BDS

FATIMA NOOR
HMH

WHY?

To know your blood sugar levels. Your diet, medicines, physical activity, sickness and stress can affect your blood sugar levels.

Knowing your blood sugar levels will help you control your diabetes. This information will also help you and your doctor or health care provider make changes to your diabetes treatment plan.

What is the principle of glucometer

- The principle behind blood glucose meters is based on reactions that are analyzed by electrochemical sensors. On each strip, there are about 10 layers, including a stiff plastic base plate, and other layers containing chemicals or acting as spacers.

- . For instance, there is a layer containing two electrodes (silver or other similar metal). There also is a layer of the immobilized enzyme, glucose oxidase, and another layer containing microcrystalline potassium ferricyanide. Specifically, the reaction of interest is between glucose and glucose oxidase. The glucose in the blood sample reacts with the glucose oxidase to form gluconic acid, which then reacts with ferricyanide to form ferrocyanide. The electrode oxidizes the ferrocyanide, and this generates a current directly proportional to the glucose concentration

When to check

Type 1

2 to 4 times a day

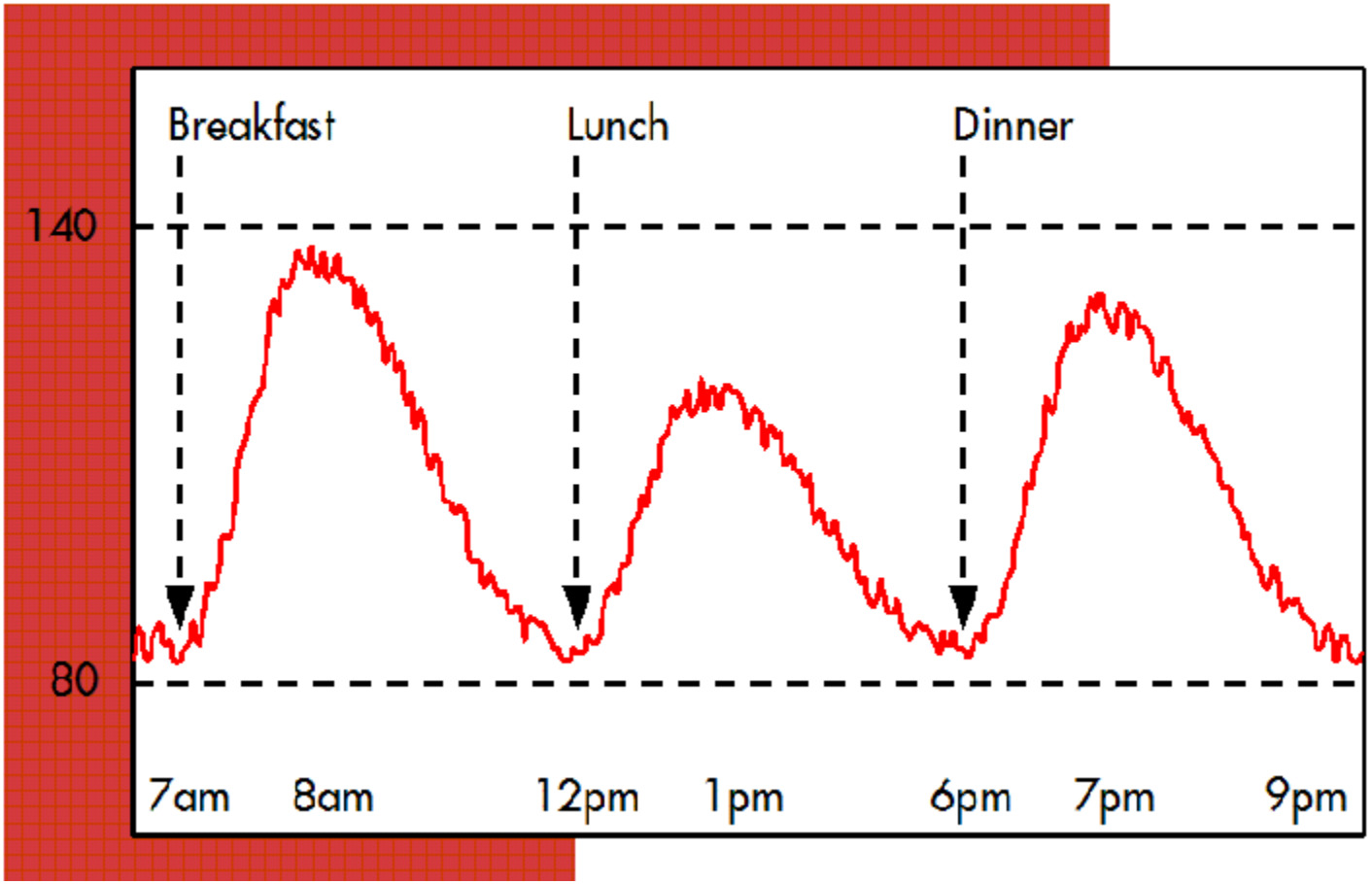
Gestational Diabetes

2 to 4 times a day

Type 2 Diabetes

Varies

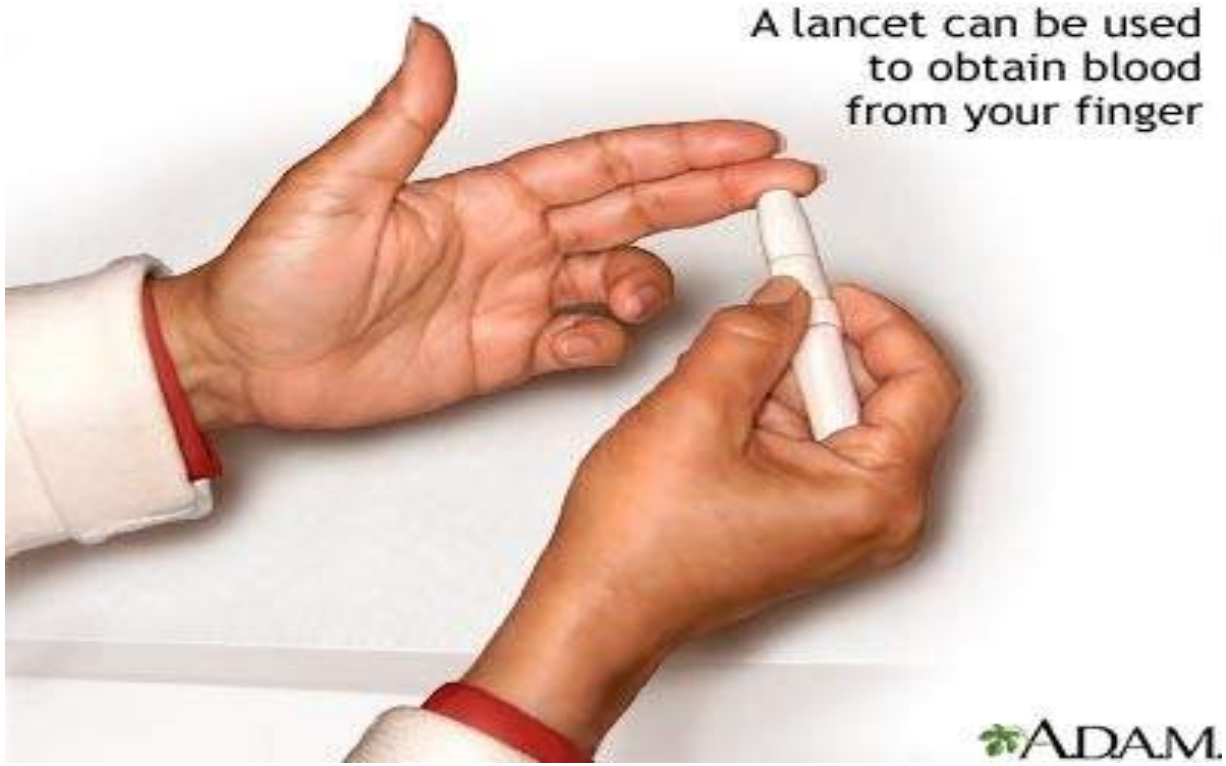
WHEN to Check



HOW?

- Know how to get a sample and How to care for your meter

Using Your Glucose Meter



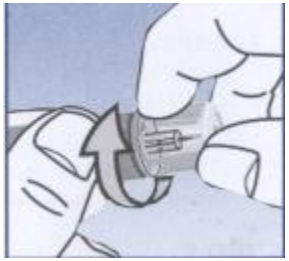

A.D.A.M.

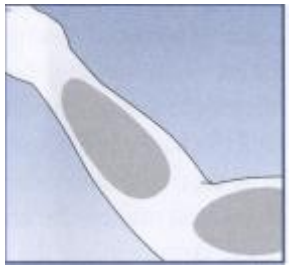

95
Blood
mg/dL
A.D.A.M.


ONETOUCH
UltraSmart
Apply Blood


95
A.D.A.M.

Alternative Site Testing

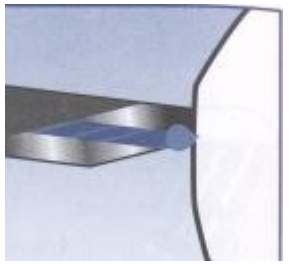








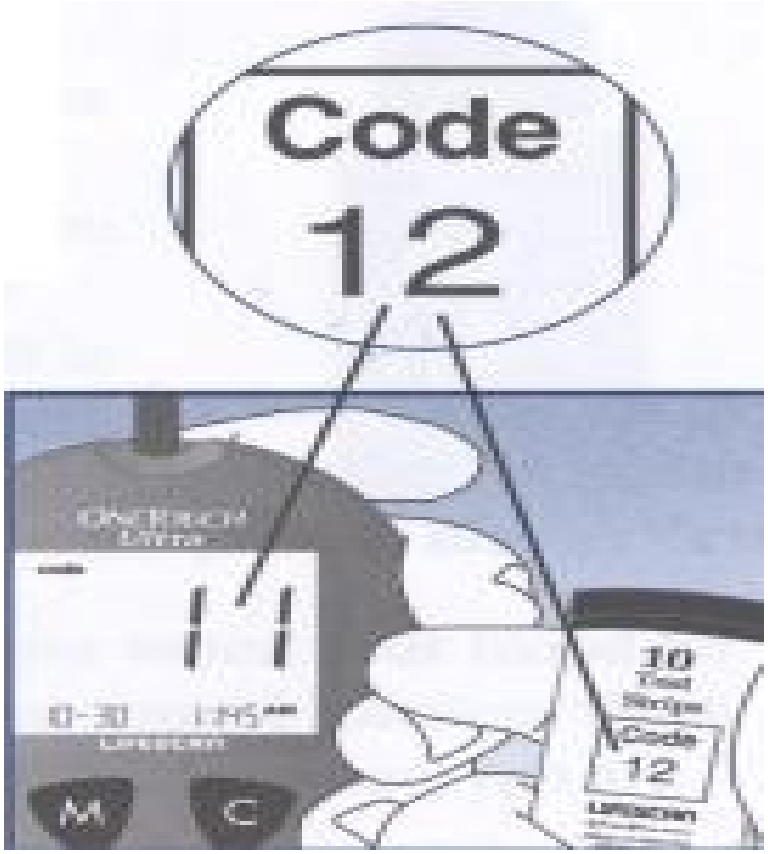




Factors That Affect Glucose Meter Performance

- ***Hematocrit***
- ***Other Substances***
- ***Altitude, Temperature and Humidity***

Calibrate the meter





Check Using "Test Quality Control Solutions" or "Electronic Controls”



- **Take Your Meter with You to The Health Care Provider's Office.**
- **Have your Health care provider or diabetes educator observe your technique**
- **Compare results to a Laboratory test done at the same time**

Important Features of Glucose Meters

Measurement Range

Whole Blood Glucose vs. Plasma Glucose

Cleaning

Display Of High And Low Glucose Values

Performance Goals for Meters

No universally accepted standards have been developed

ADA has recommended + or - 15% or 5%

International standard is being developed and recommends + or - 20 mg/dl for values under 100 and + or - 20% for higher values

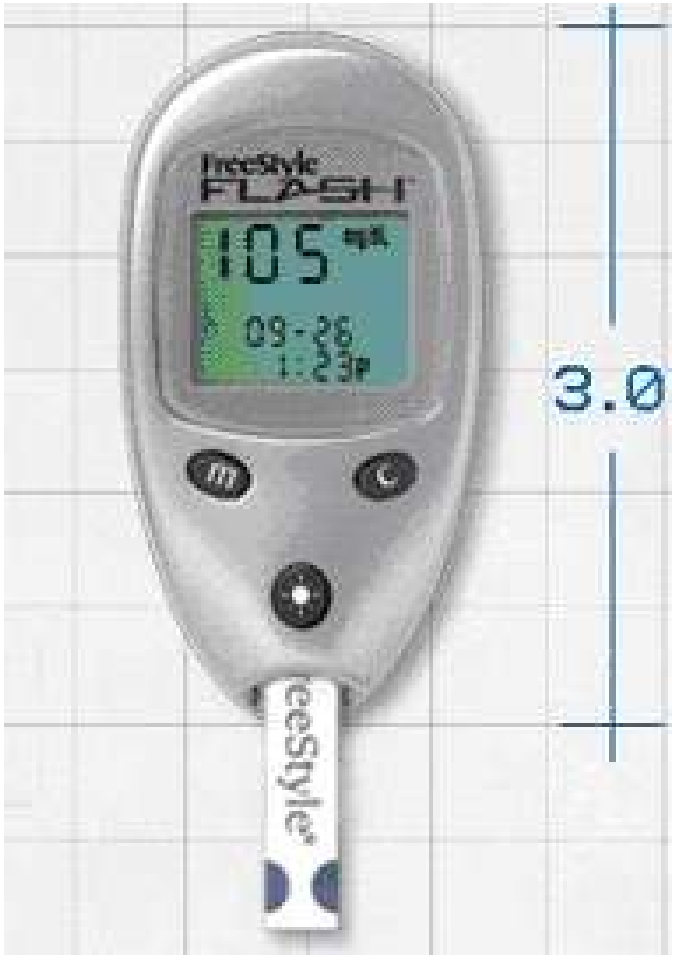
Newer meters are more accurate



This is the ultra-low price meter at Wal-Mart and

Sam's Club stores.

The FreeStyle Flash is the world's smallest blood glucose meter





Does away with handling individual test

strips



Requires only a 0.6µL (0.6 microliter) drop of blood and 15 second test time.



This disposable system is quite small and contains a meter and a vial of 50 test strips. There is no coding required. it uses 1µL (1 microliter) of blood and provides results in less than 10

seconds

Ultra and Ultra Smart







Ultra 2 meter



Flag results as before or after meal
View before and after meal averages



See the effects of food and portion choices over time
Learn to spot trends



This new meter gives reliable results in five seconds and requires one of the smallest blood samples of any meter, just 1.6µL

(1.6 microliters)



One of its major features is its Automated Drum Technology, which automates a strip-handling process with self-contained strips, calibrates the meter, and enables the test strips to be dispensed from a drum of 17 tests with a press of a

button

Minimally Invasive and Non-Invasive Glucose Meters

- MiniMed Continuous Glucose Monitoring System



- ***Cygnus GlucoWatch Biographer***

